

ENGR 102 Summer Bridge**Day 7 Student Quiz**

Ordered Validation, Priority, and Nested Decisions

20 points • approximately 20-25 minutes

Name		UIN		Section	
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Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace a priority-sensitive validation chain.**5 points**

```
quantity = -2

if quantity < 0:
    message = "invalid"
elif quantity <= 10:
    message = "small"
elif quantity <= 25:
    message = "medium"
else:
    message = "large"

print(message)
```

Name the first matching branch and explain why a negative value cannot reach small.

Question 2. Trace separate cost and classification decisions.**5 points**

```
mass = 28
purity = 90
rush = "yes"
cost = 12 + 1.8 * mass

if rush == "yes":
    cost = cost + 6

if purity < 85 or mass > 25:
    label = "hold"
else:
    label = "accepted"

print(label, cost)
```

Show cost state before and after rush, then evaluate both classification components and give exact output.

Question 3. Program construction

10 points

Write one complete Python program that satisfies every requirement.

- Read score as an integer and missing as text.
- Print invalid when score is outside 0 through 100.
- For valid input, print incomplete when missing is exactly yes, regardless of score.
- Otherwise print distinction for score at least 85, pass for score at least 70, and review for every lower valid score.
- Print exactly Result: followed by the label.

[illegible]

Test evidence / expected result